SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
Student Name:	S. Ganesh	**Reg. No.:**	1921260047
Section / Batch:		**Date:**	21/10/26

Code of Conduct S. Ganesh / 1921260047

I _____ (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula $s = \text{round}((L - 1) \times CDF)$, where $L = 256$. Determine the new intensity values and compare the contrast distribution before and after equalization.

Q2. A 3×3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3×3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

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ASSESSMENT 1 – ANALYTICAL PROBLEM SOLVING

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Q1. Histogram Equalization

Step 1: Histogram. Count each intensity: 10 occurs 2 times, 20 occurs 3 times, 30 occurs 2 times, and 40 occurs 1 time. Total pixels N = 8. **Step 2: Probability.** Divide each frequency by 8: P(10)=2/8=0.25, P(20)=3/8=0.375, P(30)=2/8=0.25, and P(40)=1/8=0.125. **Step 3: CDF.** Add probabilities cumulatively: CDF(10)=0.25, CDF(20)=0.25+0.375=0.625, CDF(30)=0.875, and CDF(40)=1.00. **Step 4: Equalization.** s=round((L−1)×CDF)=round(255×CDF). Thus 10→round(63.75)=64; 20→round(159.375)=159; 30→round(223.125)=223; and 40→255. **Conclusion:** the original values are concentrated between 10 and 40, while the new values spread across 64 to 255. This increases the available contrast range, although repeated pixels retain their frequencies.

Q2. Averaging and Median Filters

Step 1: Averaging filter. Add the nine neighbourhood values: 10+20+30+20+100+20+10+20+30=260. Divide by 9: average=260/9=28.89, approximately 29. Replace only the centre value to get [[10,20,30],[20,29,20],[10,20,30]]. **Step 2: Median filter.** Sort the values: 10,10,20,20,20,20,30,30,100. Since there are nine values, the middle (fifth) value is 20. The result is [[10,20,30],[20,20,20],[10,20,30]]. **Comparison:** averaging reduces the influence of differences by smoothing the neighbourhood, but the unusually high value 100 pulls the average upward. The median is less affected by this outlier and usually removes impulse noise while preserving edges better.

Q3. Laplacian Filter for Sharpening

Step 1: Apply the mask. The centre value 100 is multiplied by 4, giving 400. The top, bottom, left, and right neighbours are each 50 and are multiplied by −1. The four corners are multiplied by 0 and contribute nothing. **Step 2: Response.** R=400−50−50−50−50=200. **Step 3: Sharpening.** Add the response to the original centre intensity: 100+200=300. The calculated result is 300; because an 8-bit grayscale image normally allows only 0–255, it would be clipped to 255 for storage. **Conclusion:** the Laplacian responds to rapid intensity changes between a pixel and its neighbours. Adding its response strengthens transitions and makes edges and fine details more visible.

Q4. Global Thresholding

Step 1: Threshold rule. For T=100, every value greater than or equal to 100 becomes 1, and every value below 100 becomes 0. **Step 2: Convert each row.** [40,60,120] becomes [0,0,1]; [80,150,200] becomes [0,1,1]; and [30,100,180] becomes [0,1,1]. Therefore, the binary image is [[0,0,1],[0,1,1],[0,1,1]]. **Step 3: Count pixels.** There are five 1s, so foreground=5 pixels. There are four 0s, so background=4 pixels. **Conclusion:** one global threshold separates brighter pixels from darker pixels quickly. A limitation is that uneven lighting or overlapping foreground/background intensities can cause pixels to be classified incorrectly.

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ASSESSMENT 1 – ANSWERS (CONTINUED)

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Q5. Sobel Operator for Edge Detection

Step 1: Calculate Gx. Multiply corresponding pixels by the Gx mask and add: Gx=(−10+0+10)+(−20+0+180)+(−10+0+90)=240. **Step 2: Calculate Gy.** Multiply by the Gy mask and add: Gy=(−10−20−10)+(0+0+0)+(10+100+90)=160. **Step 3: Gradient magnitude.** G=√(Gx²+Gy²)=√(240²+160²)=√83200≈288.44. **Step 4: Direction.** θ=tan■¹(Gy/Gx)=tan■¹(160/240)≈33.7°. The gradient is stronger in the x-direction because |Gx|>|Gy|; the edge line is perpendicular to the gradient direction. **Conclusion:** Sobel filters measure intensity changes. Strong gradient magnitudes indicate likely edges, which can be used to locate object boundaries and support image segmentation.

Summary of final results

Question	Final result	Reg. No.
Q1	10→64; 20→159; 30→223; 40→255	1921260047
Q2	Average ≈ 29; median = 20	1921260047
Q3	Laplacian response = 200; sharpened value = 300 before clipping	1921260047
Q4	Foreground = 5; background = 4; binary image [[0,0,1],[0,1,1],[0,1,1]]	1921260047
Q5	Gx = 240; Gy = 160; magnitude ≈ 288.44; angle ≈ 33.7°	1921260047

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____